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BATTERY DIAGNOSTIC CHARGING PROTOCOL

Abstract

Systems and methods described herein relate to implementing measuring battery performance during charging. In one embodiment, a method includes receiving a diagnostic charging protocol and a set of testing points, restoring energy to a battery when the battery is determined not to be within the set of testing points, ceasing restoring energy and testing the battery when the battery is determined to be within the set of testing points, and determining battery model parameters based on battery measurements at a subset of testing points.

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Background/Summary

TECHNICAL FIELD

[0001] The subject matter described herein relates, in general, to strategies for diagnosing battery performance during charging.

BACKGROUND

[0002] The value of an electric vehicle is often determined by the health of its electric vehicle battery. Typically, the health of a vehicle battery is determined by analyzing the performance of the battery during a discharge cycle. For example, measurements of voltage and current during discharge may be used to evaluate the health of the battery.

SUMMARY

[0003] In one embodiment, a compensation measure system is disclosed. The vehicle management system includes one or more processors and a memory communicably coupled to the one or more processors. The memory stores a command module including instructions that when executed by the one or more processors cause the one or more processors to receive a diagnostic charging protocol and a set of testing points, restore energy to a battery when the battery is determined not to be within the set of testing points, cease restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points, and determine battery model parameters based on battery measurements at a subset of testing points.

[0004] In one embodiment, a non-transitory computer-readable medium including instructions that when executed by one or more processors cause the one or more processors to perform one or more functions is disclosed. The instructions include instructions to receive a diagnostic charging protocol and a set of testing points, restore energy to a battery when the battery is determined not to be within the set of testing points, cease restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points, and determine battery model parameters based on battery measurements at a subset of testing points.

[0005] In one embodiment, a method is disclosed. In one embodiment, the method includes receiving a diagnostic charging protocol and a set of testing points, restoring energy to a battery when the battery is determined not to be within the set of testing points, ceasing restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points, and determining battery model parameters based on battery measurements at a subset of testing points.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

[0007] FIG. 1 illustrates one embodiment of a vehicle within which systems and methods disclosed herein may be implemented.

[0008] FIG. 2 illustrates one embodiment of a battery diagnostic system that is associated with diagnosing battery performance.

[0009] FIG. 3 illustrates one embodiment of a cloud computing environment within which the systems and methods described herein may operate.

[0010] FIG. 4 illustrates one example of an RC equivalent battery model.

[0011] FIG. 5 illustrates one example of an HPPC test.

[0012] FIG. 6 illustrates one example of a traditional charging cycle.

[0013] FIG. 7 illustrates one example of a charging cycle using a battery diagnostic charging protocol.

[0014] FIG. 8 illustrates one example of a method for diagnosing battery performance.

DETAILED DESCRIPTION

[0015] Systems, methods, and other embodiments associated with measuring battery performance are described herein. With respect to estimating the battery performance of an electric vehicle, traditional methods focus on the performance of the battery during discharge. For example, measurements of battery voltage and current may be used to determine an estimate of battery health. In addition, charging of electric vehicle batteries is often a rather straightforward process, typically involving a charger using constant current charging to a desired voltage, after which constant potential charging is used.

[0016] An improvement to these traditional approaches is the use of a battery diagnostic charging protocol as described herein, which may be used with a battery diagnostic system. With a battery diagnostic charging protocol, both charging and diagnostic testing to determine an estimate of battery performance may be performed during the charge cycle. In such an approach, pauses in charging may be undertaken at selected testing points to perform battery tests, after which charging may resume.

[0017] Referring to FIG. 1, an example of a vehicle **100** is illustrated. As used herein, a “vehicle” is any form of motorized transport. In one or more implementations, vehicle **100** is an automobile. While arrangements will be described herein with respect to automobiles, it will be understood that embodiments are not limited to automobiles. In some implementations, vehicle **100** may be any robotic device or form of motorized transport that, for example, includes sensors to perceive aspects of the surrounding environment, and thus benefits from the functionality discussed herein associated with diagnostic charging strategies. As a further note, this disclosure generally discusses vehicle **100** as traveling on a roadway with surrounding vehicles, which are intended to be construed in a similar manner as vehicle **100** itself. That is, the surrounding vehicles may include any vehicle that may be encountered on a roadway by vehicle **100**.

[0018] Vehicle **100** also includes various elements. It will be understood that in various embodiments it may not be necessary for vehicle **100** to have all of the elements shown in FIG. 1. Vehicle **100** may have any combination of the various elements shown in FIG. 1. Further, vehicle **100** may have additional elements to those shown in FIG. 1. In some arrangements, vehicle **100** may be implemented without one or more of the elements shown in FIG. 1. While the various elements are shown as being located within vehicle **100** in FIG. 1, it will be understood that one or more of these elements may be located external to vehicle **100**. Further, the elements shown may be physically separated by large distances. For example, as discussed, one or more components of the disclosed system may be implemented within a vehicle while further (01116479) 3 components of the system are implemented within a cloud-computing environment or other system that is remote from vehicle **100**.

[0019] Some of the possible elements of vehicle **100** are shown in FIG. 1 and will be described along with subsequent figures. However, a description of many of the elements in FIG. 1 will be provided after the discussion of FIGS. 2-8 for purposes of brevity of this description. Additionally, it will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various combinations of these elements. In either case, vehicle **100** includes a battery diagnostic system **170** that is

implemented to perform methods and other functions as disclosed herein relating to implementing vehicle seat force tracking. As will be discussed in greater detail subsequently, battery diagnostic system **170**, in various embodiments, is implemented partially within vehicle **100** and as a cloud-based service. For example, in one approach, functionality associated with at least one module of battery diagnostic system **170** is implemented within vehicle **100** while further functionality is implemented within a cloud-based computing system.

[0020] With reference to FIG. 2, one embodiment of battery diagnostic system **170** of FIG. 1 is further illustrated. Battery diagnostic system **170** is shown as including processor(s) **110** from vehicle **100** of FIG. 1. Accordingly, processor(s) **110** may be a part of battery diagnostic system **170**, battery diagnostic system **170** may include a separate processor from processor **110** (s) of vehicle **100**, or battery diagnostic system **170** may access processor **110** (s) through a data bus or another communication path. In one embodiment, battery diagnostic system **170** includes memory **210**, which stores detection module **220** and command module **230**. Memory **210** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing detection module **220** and command module **230**. Detection module **220** and command module **230** are, for example, computer-readable instructions that when executed by processor(s) **110** cause processor(s) **110** to perform the various functions disclosed herein.

[0021] Battery diagnostic system **170** as illustrated in FIG. 2 is generally an abstracted form of battery diagnostic system **170** as may be implemented between vehicle **100** and a cloud-computing environment. Accordingly, battery diagnostic system **170** may be embodied at least in part within a cloud-computing environment to perform the methods described herein.

[0022] With reference to FIG. 2, detection module **220** generally includes instructions that function to control processor(s) **110** to receive data inputs from one or more sensors of vehicle **100**. The inputs are, in one embodiment, observations of one or more objects in an environment proximate to vehicle **100**, other aspects about the surroundings, or both. As provided for herein, detection module **220**, in one embodiment, acquires sensor data **250** that includes at least camera images. In further arrangements, detection module **220** acquires sensor data **250** from further sensors such as radar **123**, LiDAR **124**, and other sensors as may be suitable for identifying vehicles, locations of the vehicles, lane markers, crosswalks, traffic signs, vehicle parking areas, road surface types, curbs, vehicle barriers, and so on. In one embodiment, detection module **220** may also acquire sensor data **250** from one or more sensors that allow for implementing vehicle seat force tracking.

[0023] Accordingly, detection module **220**, in one embodiment, controls the respective sensors to provide sensor data **250**. Additionally, while detection module **220** is discussed as controlling the various sensors to provide sensor data **250**, in one or more embodiments, detection module **220** may employ other techniques to acquire sensor data **250** that are either active or passive. For example, detection module **220** may passively sniff sensor data **250** from a stream of electronic information provided by the various sensors to further components within vehicle **100**. Moreover, detection module **220** may undertake various approaches to fuse data from multiple sensors when providing sensor data **250**, from sensor data acquired over a wireless communication link (e.g., v2v) from one or more of the surrounding vehicles, or from a combination thereof. Thus, sensor data **250**, in one embodiment, represents a combination of perceptions acquired from multiple sensors.

[0024] In addition to locations of surrounding vehicles, sensor data **250** may also include, for example, odometry information, GPS data, or other location data. Moreover, detection module **220**, in one embodiment, controls the sensors to acquire sensor data about an area that encompasses 360 degrees about vehicle **100**, which may then be stored in sensor data **250**. In some embodiments, such area sensor data may be used to provide a comprehensive assessment of the surrounding environment around vehicle **100**. Of course, in alternative embodiments, detection module **220** may acquire the sensor data about a forward direction alone when, for example, vehicle **100** is not

equipped with further sensors to include additional regions about the vehicle or the additional regions are not scanned due to other reasons (e.g., unnecessary due to known current conditions).

[0025] Moreover, in one embodiment, battery diagnostic system **170** includes a database **240**. Database **240** is, in one embodiment, an electronic data structure stored in memory **210** or another data store and that is configured with routines that may be executed by processor(s) **110** for analyzing stored data, providing stored data, organizing stored data, and so on. Thus, in one embodiment, database **240** stores data used by the detection module **220** and command module **230** in executing various functions. In one embodiment, database **240** includes sensor data **250** along with, for example, metadata that characterize various aspects of sensor data **250**. For example, the metadata may include location coordinates (e.g., longitude and latitude), relative map coordinates or tile identifiers, time/date stamps from when separate sensor data **250** was generated, and so on.

[0026] Detection module **220**, in one embodiment, is further configured to perform additional tasks beyond controlling the respective sensors to acquire and provide sensor data **250**. For example, detection module **220** includes instructions that may cause processor(s) **110** to obtain battery measurements as described herein. In some embodiments, detection module **220** may receive and store battery measurements.

[0027] In one embodiment, command module **230** generally includes instructions that function to control the processor(s) **110** or collection of processors in the cloud-computing environment **300** as shown in FIG. 3.

[0028] With reference to FIG. 3, vehicle **100** may be connected to a network **305**, which allows for communication between vehicle **100** and cloud servers (e.g., cloud server **310**), infrastructure devices (e.g., infrastructure device **340**), other vehicles (e.g., vehicle **380**), and any other systems connected to network **305**. With respect to network **305**, such a network may use any form of communication or networking to exchange data, including but not limited to the Internet, Directed Short Range Communication (DSRC) service, LTE, 5G, millimeter wave (mmWave) communications, and so on.

[0029] Cloud server **310** is shown as including a processor **315** that may be a part of battery diagnostic system **170** through network **305** via communication unit **335**. In one embodiment, cloud server **310** includes a memory **320** that stores a communication module **325**. Memory **320** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing communication module **325**. Communication module **325** is, for example, computer-readable instructions that when executed by processor **315** causes processor **315** to perform the various functions disclosed herein. Moreover, in one embodiment, cloud server **310** includes database **330**. Database **330** is, in one embodiment, an electronic data structure stored in a memory **320** or another data store and that is configured with routines that may be executed by processor **315** for analyzing stored data, providing stored data, organizing stored data, and so on.

[0030] Infrastructure device **340** is shown as including a processor **345** that may be a part of battery diagnostic system **170** through network **305** via communication unit **370**. In one embodiment, infrastructure device **340** includes a memory **350** that stores a communication module **355**. Memory **350** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing communication module **355**. Communication module **355** is, for example, computer-readable instructions that when executed by processor **345** causes processor **345** to perform the various functions disclosed herein. Moreover, in one embodiment, infrastructure device **340** includes a database **360**. Database **360** is, in one embodiment, an electronic data structure stored in memory **350** or another data store and that is configured with routines that may be executed by processor **345** for analyzing stored data, providing stored data, organizing stored data, and so on.

[0031] Accordingly, in addition to information obtained from sensor data **250**, battery diagnostic system **170** may obtain information from cloud servers (e.g., cloud server **310**), infrastructure devices (e.g., infrastructure device **340**), other vehicles (e.g., vehicle **380**), and any other systems

connected to network **305**. For example, cloud servers (e.g., cloud server **310**) may be used to perform the same tasks as described herein with respect to command module **230**. For example, infrastructure device **340** may detect within its coverage area the number of vehicles awaiting charging, such that infrastructure device **340** may determine that it should not allow charging vehicles to perform the diagnostic charging protocol if the number of vehicles is above a pre-determined threshold.

[0032] The performance of a rechargeable battery, such as a lithium-ion or sodium-ion battery, is typically governed by its internal resistance, which may change with respect to temperature, state of charge, state of health, or other factors. Given this variation in internal resistance, a battery charger typically takes either a constant current or constant potential approach to charging rechargeable batteries. With respect to the constant current mode, a constant current may be used to charge the rechargeable battery, such that the changes in the internal resistance of the battery cause the charging voltage to vary. With respect to the constant potential mode, a constant voltage may be used to charge the rechargeable battery, such that the changes in the internal resistance of the battery cause the charging current to vary. Accordingly, depending on which mode is used, at least one parameter will vary in accordance with the changing internal resistance of the battery. The measurement of such a varying parameter may then allow for estimating the internal resistance of a battery according to an equivalent circuit model.

[0033] With respect to FIG. **4**, an example of an RC equivalent circuit model for a rechargeable battery is shown. In such a second-order RC equivalent circuit model, which consists of a voltage source, a resistor, and two RC networks in series: U_{oc} may denote the battery open-circuit voltage, R_0 may denote the battery ohmic resistance, R_1 and R_2 may respectively denote the charge transfer resistor and polarization resistor, and C_1 and C_2 may respectively denote the charge transfer capacitor and polarization capacitor. Furthermore, τ_1 may denote the time constant for the first RC network where $\tau_1 = R_1 C_1$ and τ_2 may denote the time constant for the second RC link where $\tau_2 = R_2 C_2$. Finally, I may denote the battery current, U may denote the battery voltage, U_1 and U_2 may respectively denote the terminal voltages of the two RC networks.

[0034] According to Kirchhoff's law, a basic relationship between the above variables may be given as follows:

$$[00001] \quad U(t) = U_{OC} - IR_0 - U_1 - U_2, \quad \text{Equation(1)}$$

$$I = \frac{U_1}{R_1} + C_1 \frac{dU_1}{dt} = \frac{U_2}{R_2} + C_2 \frac{dU_2}{dt}. \quad \text{Equation(2)}$$

[0035] With respect to FIG. **5**, an example of a voltage and current curves for a rechargeable battery under a hybrid pulse power characterization ("HPPC") test is shown. In view of the values at positions A through H of the voltage curve, the ohmic resistance R_0 may be obtained as follows:

$$[00002] \quad R_{0, \text{discharge}} = \frac{(U_A - U_B) + (U_D - U_C)}{2I}, \quad \text{Equation(3)}$$

$$R_{0, \text{charge}} = \frac{(U_F - U_E) + (U_G - U_H)}{2I}, \quad \text{Equation(4)} \quad R_{0, \text{average}} = \frac{R_{0, \text{discharge}} + R_{0, \text{charge}}}{2}, \quad \text{Equation(5)}$$

where $U_{sub.n}$ denotes the voltage measurement at position $n=A, \dots, H$ and I denotes the battery current.

[0036] Based on Equation (1) and Equation (2), the following may be obtained:

$$[00003] \quad U(t) = U_{OC} - IR_0 - U_1(t) - U_2(t), \quad \text{Equation(6)}$$

$$U_1(t) = U_1(0)e^{-\frac{t}{\tau_1}} + IR_1(1 - e^{-\frac{t}{\tau_1}}), \quad \text{Equation(7)}$$

$$U_2(t) = U_2(0)e^{-\frac{t}{\tau_2}} + IR_2(1 - e^{-\frac{t}{\tau_2}}), \quad \text{Equation(8)}$$

where $U_{sub.1}(0)$ and $U_{sub.2}(0)$ are the initial voltages at both ends of the polarization capacitor in each stage.

[0037] Segment BC is the zero-state response of RC networks to the change in input current and the voltages at both ends of the polarization capacitors may be obtained as follows:

[00004] $U_1(t) = IR_1(1 - e^{-\frac{t}{\tau_1}})$, Equation(9) $U_2(t) = IR_2(1 - e^{-\frac{t}{\tau_2}})$, Equation(10)

[0038] For the segment CD the polarization voltage can be seen as remaining unchanged until after point D of the voltage curve, such that the initial voltage at both ends of the polarization capacitors may be obtained as follows:

[00005] $U_1(0) = IR_1(1 - e^{-\frac{T}{\tau_1}})$, Equation(11) $U_2(0) = IR_2(1 - e^{-\frac{T}{\tau_2}})$, Equation(12)

where T is the pulse duration of segment BC (e.g., 20 s).

[0039] For the segment DE the input current is 0, which can be used as a zero-input response to obtain the battery voltage as follows:

[00006] $U(t) = U_{OC} - U_1(0)e^{-\frac{t}{\tau_1}} - U_2(0)e^{-\frac{t}{\tau_2}}$, Equation(13)

[0040] Accordingly, the values of $U_{sub.1}(0)$, $U_{sub.2}(0)$, $\tau_{sub.1}$, and $\tau_{sub.2}$ may be obtained by fitting a curve of Equation (13) to voltage measurements over the segment DE. Once a fit is obtained, values of $R_{sub.1}$, $R_{sub.2}$, $C_{sub.1}$, and $C_{sub.2}$ for the discharge cycle may be obtained as follows:

[00007] $R_1 = \frac{U_1(0)}{I(1 - e^{-\frac{T}{\tau_1}})}$, Equation(14) $R_2 = \frac{U_2(0)}{I(1 - e^{-\frac{T}{\tau_2}})}$, Equation(15) $C_1 = \frac{1}{R_1}$, Equation(16)

$C_2 = \frac{1}{R_2}$. Equation(17)

[0041] The process described above may be repeated with respect to segment HI to obtain $R_{sub.1}$, $R_{sub.2}$, $C_{sub.1}$, and $C_{sub.2}$ for the charge cycle. Average values may then be obtained for $R_{sub.1}$, $R_{sub.2}$, $C_{sub.1}$, and $C_{sub.2}$ based on the values obtained with respect to both the charge cycle and discharge cycle.

[0042] It should be noted that while the above is one example of using an RC equivalent model, other battery models may be also used such as higher-order RC equivalent models. In addition, it should be noted that while the above is also one example of using an HPPC test to obtain the parameters of an RC equivalent model, other methods for obtaining battery parameters may also be used.

[0043] In some embodiments, command module **230** may receive a diagnostic charging protocol. For example, a diagnostic charging protocol may instruct vehicle **100** when to cease charging and then apply a test to obtain parameters for a battery model, such as an RC equivalent model, at multiple testing points. For instance, a vehicle may be instructed to pause charging every ten minutes, when the estimated state of charge increases by 10%, or other metrics such as length of time (e.g., every 30 minutes). During each pause, command module **230** may be instructed to use a battery test, such as an HPPC test to obtain model parameters. For example, each pause may be initiated with a period of rest to achieve battery equilibration, followed by a discharge pulse at a high current for a short period, followed by a relaxation period, followed by a charge pulse at high current for a short period, and then followed by another relaxation period, after which charging may resume. In some embodiments, such a battery test may be repeated, such as by using different timing between pulses or different levels of discharge/charge pulses within the battery test (e.g., subsequent discharge/charge pulses may increase or decrease in terms of magnitude, frequency, or both). In some embodiments, the length of the relaxation periods and charging/discharging periods may also be varied according to the time it requires the battery to reach equilibration. As part of the diagnostic charging protocol, command module **230** may record the battery voltage, battery current, and other battery parameters during battery tests.

[0044] With respect to FIG. 6, an example of a traditional charging cycle is shown, where the charger begins and stays in a constant current charging until it reaches a threshold where the charger switches to constant voltage charging. With respect to FIG. 7, an example of a charging cycle using a diagnostic charging protocol is shown, where pauses are taken at testing points to

implement battery tests. It should be understood that while FIG. 7 only shows a battery test involving a rest period to achieve battery equilibrium, other battery tests may be implemented at testing points such as an HPPC test or others.

[0045] In some embodiments, command module **230** may be prompted to start charging vehicle **100** according to a diagnostic charging protocol. For example, a battery may be substantially depleted and automatically employ the diagnostic charging protocol during vehicle charging. As another example, command module **230** may receive an instruction to employ the diagnostic charging protocol during vehicle charging. In some embodiments, a vehicle operator must approve or initiate the use of the diagnostic charging protocol before it is implemented during vehicle charging. For example, the vehicle operator may be informed that the diagnostic charging protocol will add 10 minutes in charging to a desired target, such as 80% capacity, which may be rejected by the vehicle operator (e.g., because they are in a hurry). In some embodiments, a vehicle may be instructed that it cannot implement the diagnostic charging protocol, such as because a vehicle charging system does not support the diagnostic charging protocol, is a public charger, or other limitations (e.g., day/time restrictions, number of vehicles waiting to charge exceeds threshold).

[0046] In some embodiments, command module **230** may receive battery measurements, such as battery voltage, battery current, battery temperature, etc., that were obtained during a battery test. Based on the battery measurements, command module **230** may apply curve fitting, machine learning, or other techniques to determine the parameters of a battery model, such as an RC equivalent model, that matches the battery measurements. In some embodiments, command module **230** may require that the parameters of the battery model achieve a match with the battery measurements within one or more conditions (e.g., better than 99% match; standard deviation of error below a certain threshold). If command module **230** finds that a satisfactory match cannot be obtained, command module **230** may determine that the battery model is invalid.

[0047] In some embodiments, command module **230** may perform analysis with respect to one or more battery models at one or more testing points. For example, at each testing point at a different state of charge (e.g., 10%, 20%, 30%), command module **230** may have a set of model parameters associated with each state of charge, which command module may then use to determine estimates of battery characteristics, such as anode or cathode degradation, state of health, fast charging capability, or other metrics.

[0048] For example, command module **230** may use resistance values of one or more battery models at low states of charge (e.g., under 25%) to determine an estimate of anode degradation of a battery. Similarly, command module **230** may use resistance values of one or more battery models at a high states of charge (e.g., over 75%) to determine an estimate of cathode degradation of a battery. In such an embodiment, command module **230** may receive a prediction model **260** trained to provide estimates of anode degradation, cathode degradation, or both based on resistance values at different testing points (e.g., different states of charge). Such a model may be trained by test data obtained from a set of test batteries used to evaluate anode degradation, cathode degradation, or a combination of both.

[0049] As another example, command module **230** may use resistance or capacitive values of one or more battery models at different states of charge and different discharge/charge current levels to determine a capability for fast vehicle charging. In such an embodiment, command module **230** may receive a prediction model **260** trained to provide estimates of capability for fast vehicle charging based on resistance or capacitive values at different testing points (e.g., different states of charge and different discharge/charge current levels). Such a model may be trained by test data obtained from a set of test batteries used to evaluate capability for fast vehicle charging. As such, changes in resistance or capacitive values of battery models due to different states of charge, different discharge/charge current levels, or both may be used to determine a capability for fast vehicle charging, such as an estimate of remaining possible fast charges, restrictions on fast charging (e.g., not to exceed a voltage or amperage; fast charging not possible), and so on.

[0050] In some embodiments, command module **230** may adjust testing points between vehicle charges. For example, vehicle **100** during a trip may charge from 5% to 35% at a first charger, 19% to 65% at a second charger, and then 42% to full capacity at a third charger. In a situation where it is desired to achieve battery tests at every 10% of capacity, command module **230** may omit performing battery tests at 20% and 30% of capacity at the second charger as it has obtained those tests at the first charger and also omit performing battery tests at 50% and 60% of capacity at the third charger as it has obtained those tests at the second charger.

[0051] In some embodiments, command module **230** may adjust testing points based on battery history, driver profile, number of fast charges performed, or other factors. For example, as the number of fast charges increase above one or more pre-determined thresholds, command module **230** may adjust testing points to focus on higher states of charge to evaluate battery degradation due to fast charging. As another example, driver behavior may indicate to command module **230** when selecting testing points would be optimal (e.g., due to different patterns in charging behaviors between drivers).

[0052] In some embodiments, command module **230** may use resistance or capacitive values of one or more battery models at different states of charge and different discharge/charge current levels to determine a composite battery score. In such an embodiment, command module **230** may receive a prediction model **260** trained to provide a composite battery score that allows for comparing batteries on a pre-determined scale (e.g., **0** to **100**) based on resistance or capacitive values at different testing points (e.g., different states of charge and different discharge/charge current levels). Such a model may be trained by test data obtained from a set of test batteries used to determine composite battery scores. Accordingly changes in resistance or capacitive values of battery models due to different states of charge, different discharge/charge current levels, or both may be used to determine a composite battery score.

[0053] In some embodiments, vehicle **100** may also include the ability to securely transmit battery health measurements or analysis (e.g., the composite battery score) to a desired service, such as a vehicle diagnostic station or a third-party website hosting vehicle history information.

[0054] FIG. **8** illustrates a flowchart of a method **800** that is associated with using diagnostic charging strategies. Method **800** will be discussed from the perspective of the battery diagnostic system **170** of FIGS. **1** and **2**. While method **800** is discussed in combination with the battery diagnostic system **170**, it should be appreciated that the method **800** is not limited to being implemented within battery diagnostic system **170** but is instead one example of a system that may implement method **800**.

[0055] At step **810**, command module **230** may receive a diagnostic charging protocol and a set of testing points. For example, a diagnostic charging protocol may instruct charging to be paused at each testing point and a battery test to be performed once the battery reaches equilibrium. In some embodiments, the set of testing points may be pre-determined, while in other embodiments the testing points may be adjusted or altered. In some embodiments, the adjustment or alteration of the testing points may be performed by the battery diagnostic charging protocol.

[0056] At step **820**, command module **230** may restore energy to a battery when the battery is determined not to be within the set of testing points. For example, when the battery is not a testing point, it may be charged the same as a battery not undergoing testing.

[0057] At step **830**, command module **230** may cease restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points. For example, if the battery is at a testing point, charging may be paused to allow the battery to reach equilibrium, after which a battery test, such as an HPPC test, is performed. After the battery test is complete, charging may once again resume.

[0058] At step **840**, command module **230** may determine battery model parameters based on battery measurements at a subset of testing points. For example, after obtaining a set of battery measurements at different testing points, command module **230** may use machine learning models

to determine battery health, the expected number of fast charging events remaining, a composite battery score, or other metrics. In some embodiments, the subset of testing points may be less than the full set of testing points (e.g., because not all testing points were tested, some testing points were invalid), while in other embodiments the subset of testing points may equal the full set of testing points (e.g., because all the testing points were encountered, data across tests was amalgamated to obtain data at all testing points).

[0059] FIG. 1 will now be discussed in full detail as an example environment within which the system and methods disclosed herein may operate. In some instances, vehicle **100** is configured to switch selectively between various modes, such as an autonomous mode, one or more semi-autonomous operational modes, a manual mode, etc. Such switching may be implemented in a suitable manner, now known, or later developed. “Manual mode” means that all of or a majority of the navigation/maneuvering of the vehicle is performed according to inputs received from a user (e.g., human driver). In one or more arrangements, vehicle **100** may be a conventional vehicle that is configured to operate in only a manual mode.

[0060] In one or more embodiments, vehicle **100** is an autonomous vehicle. As used herein, “autonomous vehicle” refers to a vehicle that operates in an autonomous mode. “Autonomous mode” refers to using one or more computing systems to control vehicle **100**, such as providing navigation/maneuvering of vehicle **100** along a travel route, with minimal or no input from a human driver. In one or more embodiments, vehicle **100** is either highly automated or completely automated. In one embodiment, vehicle **100** is configured with one or more semi-autonomous operational modes in which one or more computing systems perform a portion of the navigation/maneuvering of the vehicle along a travel route, and a vehicle operator (i.e., driver) provides inputs to the vehicle to perform a portion of the navigation/maneuvering of vehicle **100** along a travel route.

[0061] Vehicle **100** may include one or more processors **110**. In one or more arrangements, processor(s) **110** may be a main processor of vehicle **100**. For instance, processor(s) **110** may be an electronic control unit (ECU). Vehicle **100** may include one or more data stores **115** for storing one or more types of data. Data store(s) **115** may include volatile memory, non-volatile memory, or both. Examples of suitable data store(s) **115** include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. Data store(s) **115** may be a component of processor(s) **110**, or data store **115** may be operatively connected to processor(s) **110** for use thereby. The term “operatively connected,” as used throughout this description, may include direct or indirect connections, including connections without direct physical contact.

[0062] In one or more arrangements, data store(s) **115** may include map data **116**. Map data **116** may include maps of one or more geographic areas. In some instances, map data **116** may include information or data on roads, traffic control devices, road markings, structures, features, landmarks, or any combination thereof in the one or more geographic areas. Map data **116** may be in any suitable form. In some instances, map data **116** may include aerial views of an area. In some instances, map data **116** may include ground views of an area, including 360-degree ground views. Map data **116** may include measurements, dimensions, distances, information, or any combination thereof for one or more items included in map data **116**. Map data **116** may also include measurements, dimensions, distances, information, or any combination thereof relative to other items included in map data **116**. Map data **116** may include a digital map with information about road geometry. Map data **116** may be high quality, highly detailed, or both.

[0063] In one or more arrangements, map data **116** may include one or more terrain maps **117**. Terrain map(s) **117** may include information about the ground, terrain, roads, surfaces, other features, or any combination thereof of one or more geographic areas. Terrain map(s) **117** may

include elevation data in the one or more geographic areas. Terrain map(s) **117** may be high quality, highly detailed, or both. Terrain map(s) **117** may define one or more ground surfaces, which may include paved roads, unpaved roads, land, and other things that define a ground surface.

[0064] In one or more arrangements, map data **116** may include one or more static obstacle maps **118**. Static obstacle map(s) **118** may include information about one or more static obstacles located within one or more geographic areas. A “static obstacle” is a physical object whose position does not change or substantially change over a period of time and whose size does not change or substantially change over a period of time. Examples of static obstacles include trees, buildings, curbs, fences, railings, medians, utility poles, statues, monuments, signs, benches, furniture, mailboxes, large rocks, hills. The static obstacles may be objects that extend above ground level. The one or more static obstacles included in static obstacle map(s) **118** may have location data, size data, dimension data, material data, other data, or any combination thereof, associated with it. Static obstacle map(s) **118** may include measurements, dimensions, distances, information, or any combination thereof for one or more static obstacles. Static obstacle map(s) **118** may be high quality, highly detailed, or both. Static obstacle map(s) **118** may be updated to reflect changes within a mapped area.

[0065] Data store(s) **115** may include sensor data **119**. In this context, “sensor data” means any information about the sensors that vehicle **100** is equipped with, including the capabilities and other information about such sensors. As will be explained below, vehicle **100** may include sensor system **120**. Sensor data **119** may relate to one or more sensors of sensor system **120**. As an example, in one or more arrangements, sensor data **119** may include information on one or more LIDAR sensors **124** of sensor system **120**.

[0066] In some instances, at least a portion of map data **116** or sensor data **119** may be located in data stores(s) **115** located onboard vehicle **100**. Alternatively, or in addition, at least a portion of map data **116** or sensor data **119** may be located in data stores(s) **115** that are located remotely from vehicle **100**.

[0067] As noted above, vehicle **100** may include sensor system **120**. Sensor system **120** may include one or more sensors. “Sensor” means any device, component, or system that may detect or sense something. The one or more sensors may be configured to sense, detect, or perform both in real-time. As used herein, the term “real-time” means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep up with some external process.

[0068] In arrangements in which sensor system **120** includes a plurality of sensors, the sensors may work independently from each other. Alternatively, two or more of the sensors may work in combination with each other. In such an embodiment, the two or more sensors may form a sensor network. Sensor system **120**, the one or more sensors, or both may be operatively connected to processor(s) **110**, data store(s) **115**, another element of vehicle **100** (including any of the elements shown in FIG. **1**), or any combination thereof. Sensor system **120** may acquire data of at least a portion of the external environment of vehicle **100** (e.g., nearby vehicles).

[0069] Sensor system **120** may include any suitable type of sensor. Various examples of different types of sensors will be described herein. However, it will be understood that the embodiments are not limited to the particular sensors described. Sensor system **120** may include one or more vehicle sensors **121**. Vehicle sensor(s) **121** may detect, determine, sense, or acquire in a combination thereof information about vehicle **100** itself. In one or more arrangements, vehicle sensor(s) **121** may be configured to detect, sense, or acquire in a combination thereof position and orientation changes of vehicle **100**, such as, for example, based on inertial acceleration. In one or more arrangements, vehicle sensor(s) **121** may include one or more accelerometers, one or more gyroscopes, an inertial measurement unit (IMU), a dead-reckoning system, a global navigation satellite system (GNSS), a global positioning system (GPS), a navigation system **147**, other suitable sensors, or any combination thereof. Vehicle sensor(s) **121** may be configured to detect,

sense, or acquire in a combination thereof one or more characteristics of vehicle **100**. In one or more arrangements, vehicle sensor(s) **121** may include a speedometer to determine a current speed of vehicle **100**.

[0070] Alternatively, or in addition, sensor system **120** may include one or more environment sensors **122** configured to acquire, sense, or acquire in a combination thereof driving environment data. “Driving environment data” includes data or information about the external environment in which an autonomous vehicle is located or one or more portions thereof. For example, environment sensor(s) **122** may be configured to detect, quantify, sense, or acquire in any combination thereof obstacles in at least a portion of the external environment of vehicle **100**, information/data about such obstacles, or a combination thereof. Such obstacles may be comprised of stationary objects, dynamic objects, or a combination thereof. Environment sensor(s) **122** may be configured to detect, measure, quantify, sense, or acquire in any combination thereof other things in the external environment of vehicle **100**, such as, for example, lane markers, signs, traffic lights, traffic signs, lane lines, crosswalks, curbs proximate to vehicle **100**, off-road objects, etc.

[0071] Various examples of sensors of sensor system **120** will be described herein. The example sensors may be part of the one or more environment sensor(s) **122**, the one or more vehicle sensors **121**, or both. However, it will be understood that the embodiments are not limited to the particular sensors described.

[0072] As an example, in one or more arrangements, sensor system **120** may include one or more radar sensors **123**, one or more LIDAR sensors **124**, one or more sonar sensors **125**, one or more cameras **126**, or any combination thereof. In one or more arrangements, camera(s) **126** may be high dynamic range (HDR) cameras or infrared (IR) cameras.

[0073] Vehicle **100** may include an input system **130**. An “input system” includes any device, component, system, element or arrangement or groups thereof that enable information/data to be entered into a machine. Input system **130** may receive an input from a vehicle passenger (e.g., a driver or a passenger). Vehicle **100** may include an output system **135**. An “output system” includes any device, component, or arrangement or groups thereof that enable information/data to be presented to a vehicle passenger (e.g., a person, a vehicle passenger, etc.).

[0074] Vehicle **100** may include one or more vehicle systems **140**. Various examples of vehicle system(s) **140** are shown in FIG. 1. However, vehicle **100** may include more, fewer, or different vehicle systems. It should be appreciated that although particular vehicle systems are separately defined, each or any of the systems or portions thereof may be otherwise combined or segregated via hardware, software, or a combination thereof within vehicle **100**. Vehicle **100** may include a propulsion system **141**, a braking system **142**, a steering system **143**, throttle system **144**, a transmission system **145**, a signaling system **146**, a navigation system **147**, other systems, or any combination thereof. Each of these systems may include one or more devices, components, or combinations thereof, now known or later developed.

[0075] Navigation system **147** may include one or more devices, applications, or combinations thereof, now known or later developed, configured to determine the geographic location of the vehicle **100**, to determine a travel route for vehicle **100**, or to determine both. Navigation system **147** may include one or more mapping applications to determine a travel route for vehicle **100**. Navigation system **147** may include a global positioning system, a local positioning system, a geolocation system, or any combination thereof.

[0076] Processor(s) **110**, battery diagnostic system **170**, automated driving module(s) **160**, or any combination thereof may be operatively connected to communicate with various aspects of vehicle system(s) **140** or individual components thereof. For example, returning to FIG. 1, processor(s) **110**, automated driving module(s) **160**, or a combination thereof may be in communication to send or receive information from various aspects of vehicle system(s) **140** to control the movement, speed, maneuvering, heading, direction, etc. of vehicle **100**. Processor(s) **110**, battery diagnostic system **170**, automated driving module(s) **160**, or any combination thereof may control some or all

of these vehicle system(s) **140** and, thus, may be partially or fully autonomous.

[0077] Processor(s) **110**, battery diagnostic system **170**, automated driving module(s) **160**, or any combination thereof may be operable to control at least one of the navigation or maneuvering of vehicle **100** by controlling one or more of vehicle systems **140** or components thereof. For instance, when operating in an autonomous mode, processor(s) **110**, battery diagnostic system **170**, automated driving module(s) **160**, or any combination thereof may control the direction, speed, or both of vehicle **100**. Processor(s) **110**, battery diagnostic system **170**, automated driving module(s) **160**, or any combination thereof may cause vehicle **100** to accelerate (e.g., by increasing the supply of fuel provided to the engine), decelerate (e.g., by decreasing the supply of fuel to the engine, by applying brakes), change direction (e.g., by turning the front two wheels), or perform any combination thereof. As used herein, “cause” or “causing” means to make, force, compel, direct, command, instruct, enable, or in any combination thereof an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner.

[0078] Vehicle **100** may include one or more actuators **150**. Actuator(s) **150** may be any element or combination of elements operable to modify, adjust, alter, or in any combination thereof one or more of vehicle systems **140** or components thereof to responsive to receiving signals or other inputs from processor(s) **110**, automated driving module(s) **160**, or a combination thereof. Any suitable actuator may be used. For instance, actuator(s) **150** may include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and piezoelectric actuators, just to name a few possibilities.

[0079] Vehicle **100** may include one or more modules, at least some of which are described herein. The modules may be implemented as computer-readable program code that, when executed by processor(s) **110**, implement one or more of the various processes described herein. One or more of the modules may be a component of processor(s) **110**, or one or more of the modules may be executed on or distributed among other processing systems to which processor(s) **110** is operatively connected. The modules may include instructions (e.g., program logic) executable by processor(s) **110**. Alternatively, or in addition, data store(s) **115** may contain such instructions.

[0080] In one or more arrangements, one or more of the modules described herein may include artificial or computational intelligence elements, e.g., neural network, fuzzy logic, or other machine learning algorithms. Further, in one or more arrangements, one or more of the modules may be distributed among a plurality of the modules described herein. In one or more arrangements, two or more of the modules described herein may be combined into a single module.

[0081] Vehicle **100** may include one or more autonomous driving modules **160**. Automated driving module(s) **160** may be configured to receive data from sensor system **120** or any other type of system capable of capturing information relating to vehicle **100**, the external environment of the vehicle **100**, or a combination thereof. In one or more arrangements, automated driving module(s) **160** may use such data to generate one or more driving scene models. Automated driving module(s) **160** may determine position and velocity of vehicle **100**. Automated driving module(s) **160** may determine the location of obstacles, obstacles, or other environmental features including traffic signs, trees, shrubs, neighboring vehicles, pedestrians, etc.

[0082] Automated driving module(s) **160** may be configured to receive, determine, or in a combination thereof location information for obstacles within the external environment of vehicle **100**, which may be used by processor(s) **110**, one or more of the modules described herein, or any combination thereof to estimate: a position or orientation of vehicle **100**; a vehicle position or orientation in global coordinates based on signals from a plurality of satellites or other geolocation systems; or any other data/signals that could be used to determine a position or orientation of vehicle **100** with respect to its environment for use in either creating a map or determining the position of vehicle **100** in respect to map data.

[0083] Automated driving module(s) **160** either independently or in combination with battery diagnostic system **170** may be configured to determine travel path(s), current autonomous driving

maneuvers for vehicle **100**, future autonomous driving maneuvers, modifications to current autonomous driving maneuvers, etc. Such determinations by automated driving module(s) **160** may be based on data acquired by sensor system **120**, driving scene models, data from any other suitable source such as determinations from sensor data **250**, or any combination thereof. In general, automated driving module(s) **160** may function to implement different levels of automation, including advanced driving assistance (ADAS) functions, semi-autonomous functions, and fully autonomous functions. “Driving maneuver” means one or more actions that affect the movement of a vehicle. Examples of driving maneuvers include accelerating, decelerating, braking, turning, moving in a lateral direction of vehicle **100**, changing travel lanes, merging into a travel lane, and reversing, just to name a few possibilities. Automated driving module(s) **160** may be configured to implement driving maneuvers. Automated driving module(s) **160** may cause, directly or indirectly, such autonomous driving maneuvers to be implemented. As used herein, “cause” or “causing” means to make, command, instruct, enable, or in any combination thereof an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner. Automated driving module(s) **160** may be configured to execute various vehicle functions, whether individually or in combination, to transmit data to, receive data from, interact with, or to control vehicle **100** or one or more systems thereof (e.g., one or more of vehicle systems **140**).

[0084] Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. **1-8**, but the embodiments are not limited to the illustrated structure or application.

[0085] The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowcharts or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

[0086] The systems, components, or processes described above may be realized in hardware or a combination of hardware and software and may be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. Any kind of processing system or another apparatus adapted for carrying out the methods described herein is suited. A typical combination of hardware and software may be a processing system with computer-usable program code that, when being loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components, or processes also may be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes described herein. These elements also may be embedded in an application product which comprises all the features enabling the implementation of the methods described herein and, which when loaded in a processing system, is able to carry out these methods.

[0087] Furthermore, arrangements described herein may take the form of a computer program product embodied in one or more computer-readable media having computer-readable program code embodied, e.g., stored, thereon. Any combination of one or more computer-readable media

may be utilized. The computer-readable medium may be a computer-readable signal medium or a computer-readable storage medium. The phrase “computer-readable storage medium” means a non-transitory storage medium. A computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer-readable storage medium would include the following: a portable computer diskette, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that may contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0088] Generally, modules as used herein include routines, programs, objects, components, data structures, and so on that perform particular tasks or implement particular data types. In further aspects, a memory generally stores the noted modules. The memory associated with a module may be a buffer or cache embedded within a processor, a RAM, a ROM, a flash memory, or another suitable electronic storage medium. In still further aspects, a module as envisioned by the present disclosure is implemented as an application-specific integrated circuit (ASIC), a hardware component of a system on a chip (SoC), as a programmable logic array (PLA), or as another suitable hardware component that is embedded with a defined configuration set (e.g., instructions) for performing the disclosed functions.

[0089] Program code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber, cable, RF, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present arrangements may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java™, Smalltalk, C++, or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on a user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer, or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0090] The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . . ” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).

[0091] Aspects herein may be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A system, comprising: a processor; and a memory communicably coupled to the processor and storing machine-readable instructions that, when executed by the processor, cause the processor to: receive a diagnostic charging protocol and a set of testing points; restore energy to a battery when

the battery is determined not to be within the set of testing points; cease restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points; and determine battery model parameters based on battery measurements at a subset of testing points.

2. The system of claim 1, wherein the machine-readable instructions to test the battery uses different discharge/charge levels.

3. The system of claim 1, wherein the machine-readable instructions to test the battery uses hybrid pulse power characterization.

4. The system of claim 1, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: obtain an estimate of electrode degradation based on a set of battery model parameters.

5. The system of claim 1, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: obtain an estimate of a capability for fast charging based on a set of battery model parameters.

6. The system of claim 4, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: display the estimate of electrode degradation.

7. The system of claim 5, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: restrict fast charging based on if the capability for fast charging satisfies a criteria.

8. A non-transitory computer-readable medium including instructions that when executed by one or more processors cause the one or more processors to: receive a diagnostic charging protocol and a set of testing points; restore energy to a battery when the battery is determined not to be within the set of testing points; cease restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points; and determine battery model parameters based on battery measurements at a subset of testing points.

9. The non-transitory computer-readable medium of claim 8, wherein the instructions to test the battery uses different discharge/charge levels.

10. The non-transitory computer-readable medium of claim 8, wherein the instructions to test the battery uses hybrid pulse power characterization.

11. The non-transitory computer-readable medium of claim 8, wherein the instructions further include to: obtain an estimate of electrode degradation based on a set of battery model parameters.

12. The non-transitory computer-readable medium of claim 8, wherein the instructions further include to: obtain an estimate of a capability for fast charging based on a set of battery model parameters.

13. The non-transitory computer-readable medium of claim 12, wherein the instructions further include to: restrict fast charging based on if the capability for fast charging satisfies a criteria.

14. A method, comprising: receiving a diagnostic charging protocol and a set of testing points; restoring energy to a battery when the battery is determined not to be within the set of testing points; ceasing restoring energy and apply a test to the battery when the battery is determined to be within the set of testing points; and determining battery model parameters based on battery measurements at a subset of testing points.

15. The method of claim 14, wherein the test to the battery uses different discharge/charge levels.

16. The method of claim 14, wherein the test to the battery uses hybrid pulse power characterization.

17. The method of claim 14, further comprising: obtaining an estimate of electrode degradation based on a set of battery model parameters.

18. The method of claim 14, further comprising: obtaining an estimate of a capability for fast charging based on a set of battery model parameters.

19. The method of claim 17, further comprising: displaying the estimate of electrode degradation.

20. The method of claim 18, further comprising: restricting fast charging based on if the capability for fast charging satisfies a criteria.

